Statistical Fraud Detection in Financial Transactions

Using **Mahalanobis Distance** and **EWMA** for Explainable, Unsupervised Anomaly Detection

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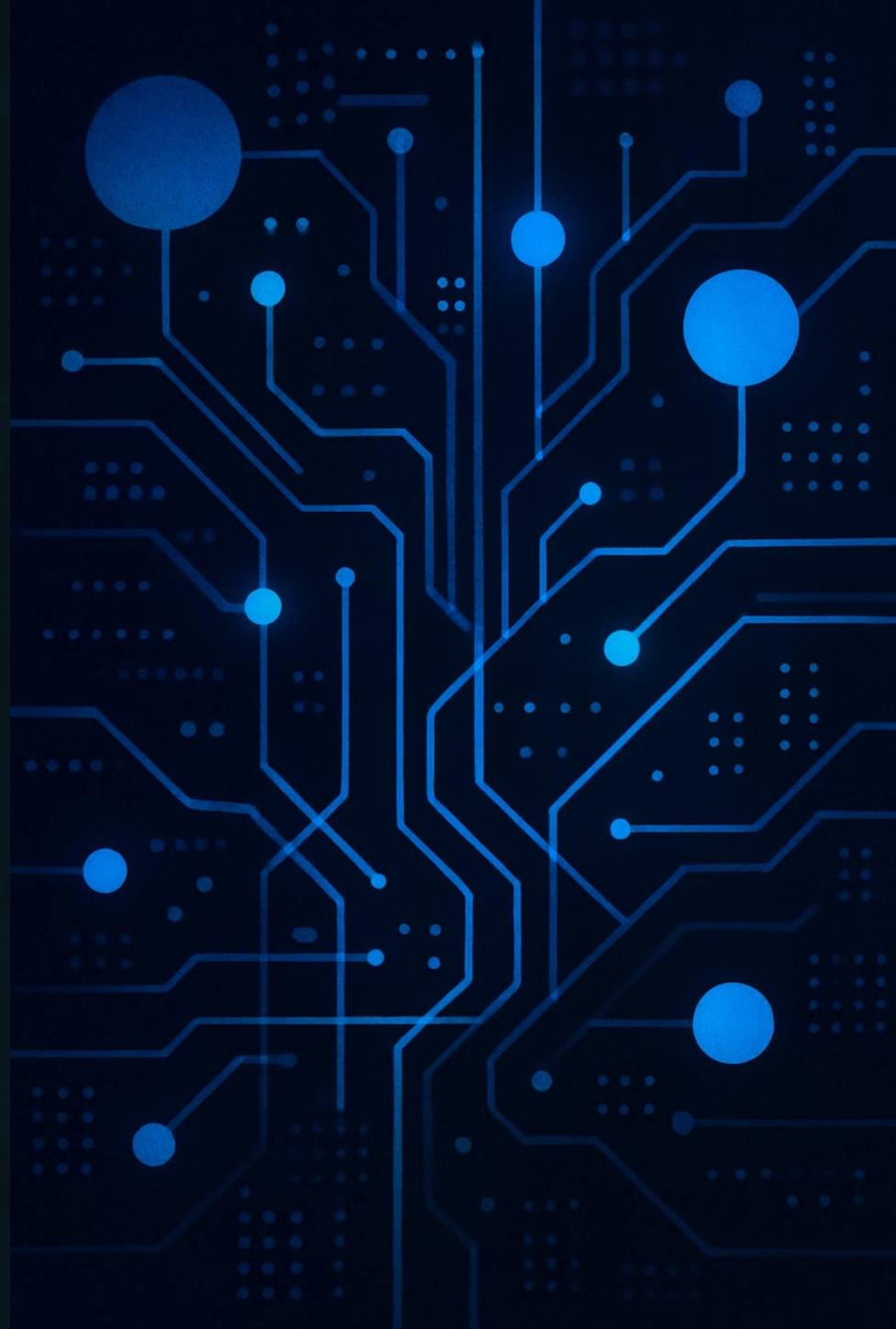


IMPLEMENTATION TOPIC

Financial Fraud: A Growing Threat

Financial fraud costs the global economy over **$5 trillion annually**. As digital banking grows, so do sophisticated attack patterns — yet most detection systems struggle to keep pace.

Scale

Billions of daily transactions make manual review impossible

Complexity

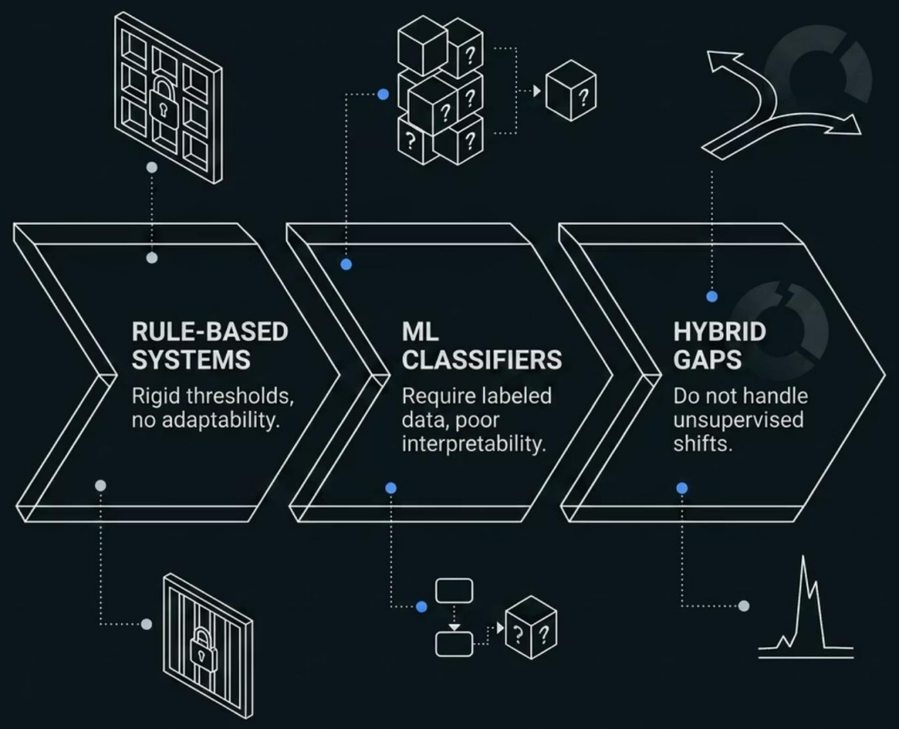
Fraud patterns evolve rapidly, bypassing static rules

Cost

False positives frustrate customers and waste resources

EXISTING SYSTEMS

Why Current Detection Falls Short



Key Limitations

- **No adaptability** — Rule engines cannot respond to evolving fraud patterns
- **Poor interpretability** — ML models act as black boxes, making audit trails difficult
- **Labeled data dependency** — Supervised models require costly, often unavailable fraud labels

Our goal: a **statistical, explainable** prototype that works without labeled data.

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PROPOSED APPROACH

A Hybrid Statistical Model

We combine two complementary statistical techniques to detect fraud from **two angles**: global deviation and temporal behavior.

Mahalanobis Distance

Measures how far a transaction deviates from the **global normal pattern** across all features simultaneously — detects outliers in multivariate space.

EWMA

Tracks **behavioral drift over time** for each account — flags sudden changes in spending patterns that static models miss.

Combined Risk Score

A weighted fusion of both signals produces a single, **interpretable risk score** — Low, Medium, or High.

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DATASET & FEATURES

Finance-Specific Synthetic Data

Raw Transaction Features

- `amount` — transaction value
- `is_international` — cross-border flag
- `failed_login_count` — security signal
- `device_change_flag` — new device indicator
- `new_beneficiary_flag` — recipient risk
- `ip_risk_score` — geolocation risk
- `amount_to_balance_ratio` — proportional spend

Engineered Features

- `time_gap_minutes` — time since last transaction
- `account_txn_count` — rolling activity volume
- `amount_deviation` — deviation from account mean
- `EWMA score` — smoothed behavioral baseline

Data simulates real banking scenarios with injected fraudulent patterns for evaluation.

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CORE METHOD — 1 OF 2

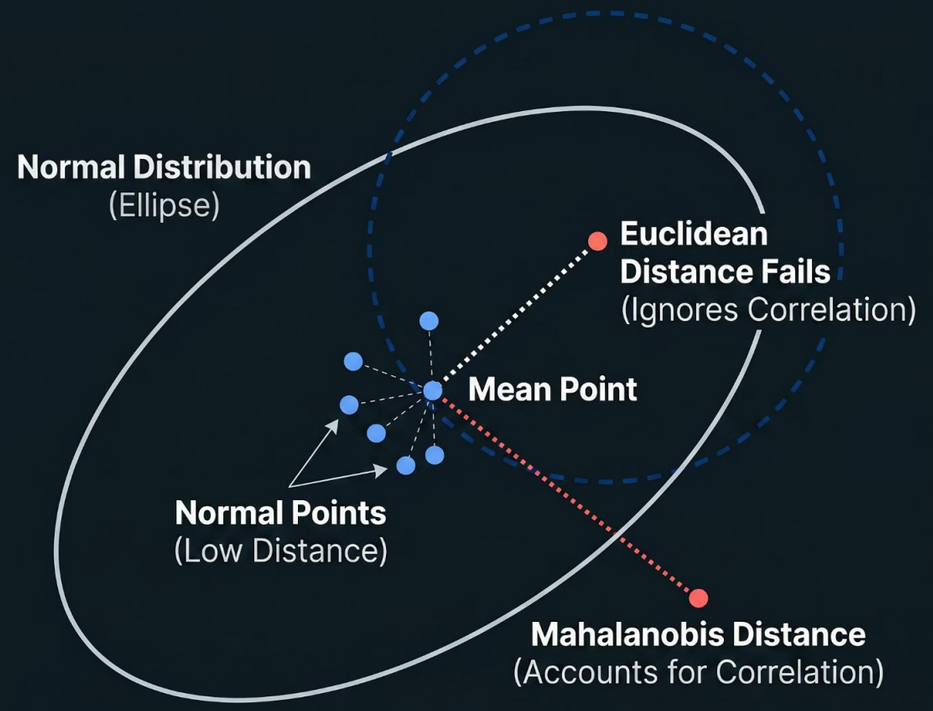
Mahalanobis Distance

Mahalanobis Distance measures how many **standard deviations** a point lies from the center of a distribution — accounting for correlations between features.

$$D(x) = \sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

- x = transaction feature vector
- μ = mean of normal transactions
- Σ^{-1} = inverse covariance matrix

A **large D(x)** indicates the transaction is statistically unusual — a potential fraud signal.



Points outside the ellipse are anomalies. Mahalanobis Distance uses correlation to accurately identify outliers, while Euclidean fails.

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CORE METHOD — 2 OF 2

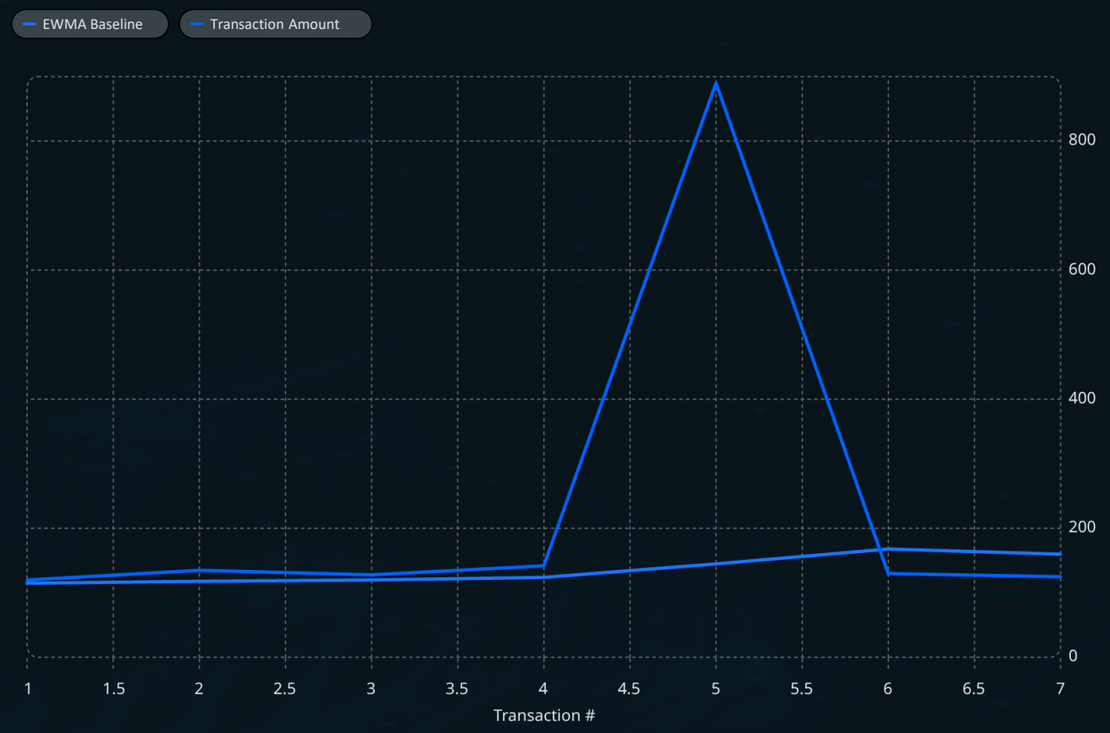
EWMA: Tracking Behavioral Drift

The **Exponentially Weighted Moving Average** gives more weight to recent transactions, creating a dynamic baseline for each account.

$$S_t = \alpha x_t + (1 - \alpha) S_{t-1}$$

- *\alpha* (0–1) controls sensitivity — higher α = more responsive
- *x_t* = current transaction feature value
- *S_{t-1}* = previous EWMA score

A sudden spike in |*x_t* - *S_t*| signals a **behavioral shift** — even if the transaction looks normal globally.



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SCORING & RESULTS

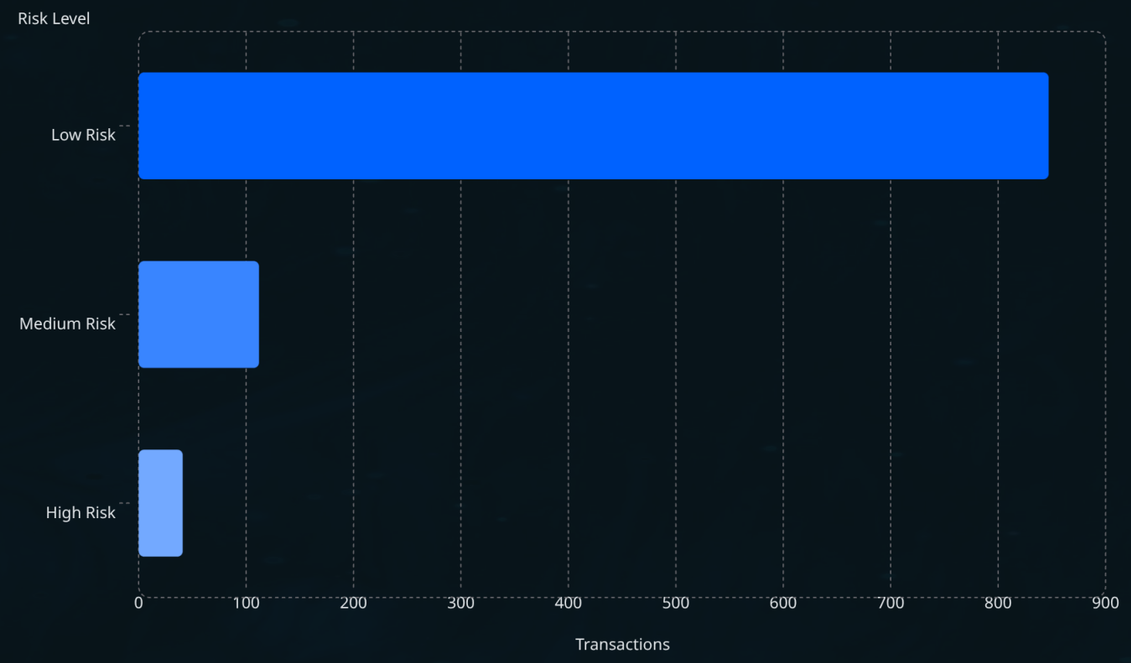
Risk Score & Detection Results

Weighted Risk Formula

Both signals are normalized to [0, 1] and combined:

$$\text{Risk} = 0.6 \times D_{\text{norm}} + 0.4 \times \text{EWMA}_{\text{norm}}$$

- **Low Risk** — Score < 0.4 → Normal
- **Medium Risk** — Score 0.4–0.7 → Review
- **High Risk** — Score > 0.7 → Flag immediately



Key Insight: Over 84% of transactions are normal — our model focuses attention on the critical 4% that are high-risk.

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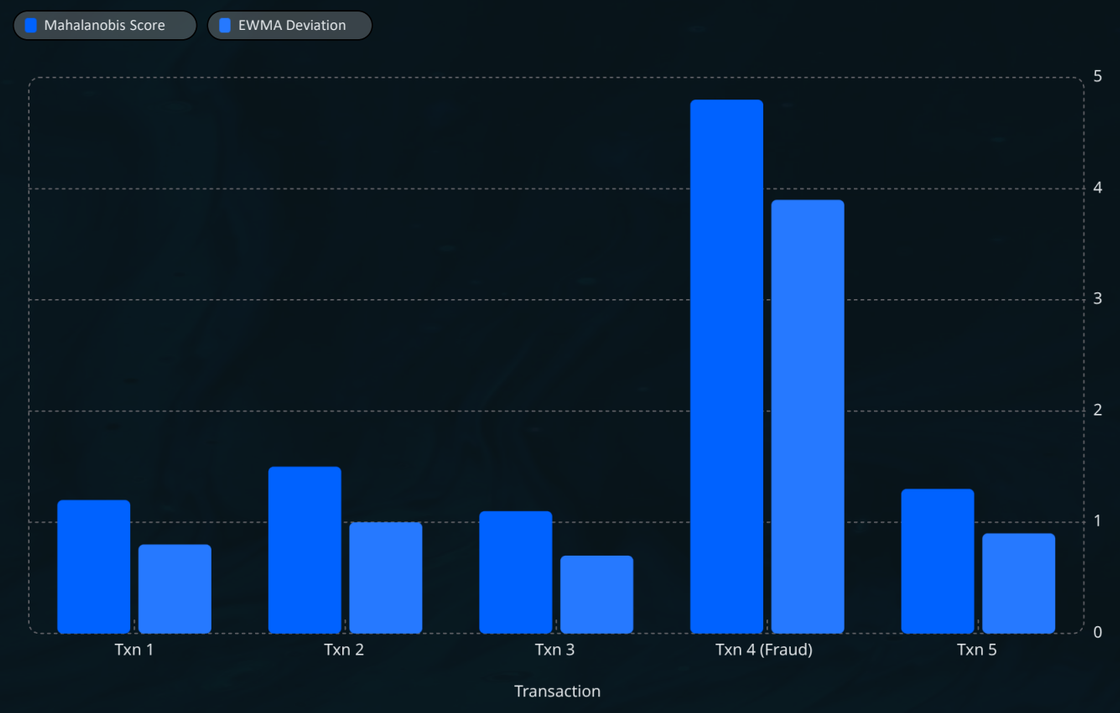
CASE STUDY

One Account, One Suspicious Spike



Interpretation

Account #1046 shows consistently low scores across four transactions. Transaction 4 triggers **both** detectors:

- **Mahalanobis** — unusual feature combination (international + new device + high amount)
- **EWMA** — amount 6× the account's recent average

Individually, each signal might be ambiguous. Together, they produce a **High Risk** classification with a clear audit trail.

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SUMMARY

Strengths, Limitations & Future Work

Advantages

- Fully **explainable** — every score is traceable
- No labeled fraud data required
- Detects both global anomalies and behavioral drift

Limitations

- Sensitive to covariance matrix quality
- Trained on synthetic data only
- Not yet a real-time streaming system

Future Improvements

→ Integrate a supervised ML classifier for hybrid detection

→ Deploy on real-time streaming platforms (e.g., Kafka)

→ Validate on real financial datasets (e.g., IEEE-CIS)

→ Improve normalization and adaptive thresholding

Conclusion: A transparent, unsupervised hybrid model that combines global anomaly detection with temporal behavior tracking — no labels required.

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CONCLUSION

Empowering Fraud Detection: A Transparent Approach

Our prototype combines **Mahalanobis Distance** and **EWMA** for explainable, unsupervised anomaly detection in financial transactions.

Transparent Methodology

Every score is traceable for auditability.

No Labeled Data Required

Detects new fraud patterns without labels.

Dual Detection Approach

Catches outliers and behavior shifts.